



Bayesian Filtering Approaches for 3D Localization of Quadruped Robots on the Grand Tour Dataset

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Introduction

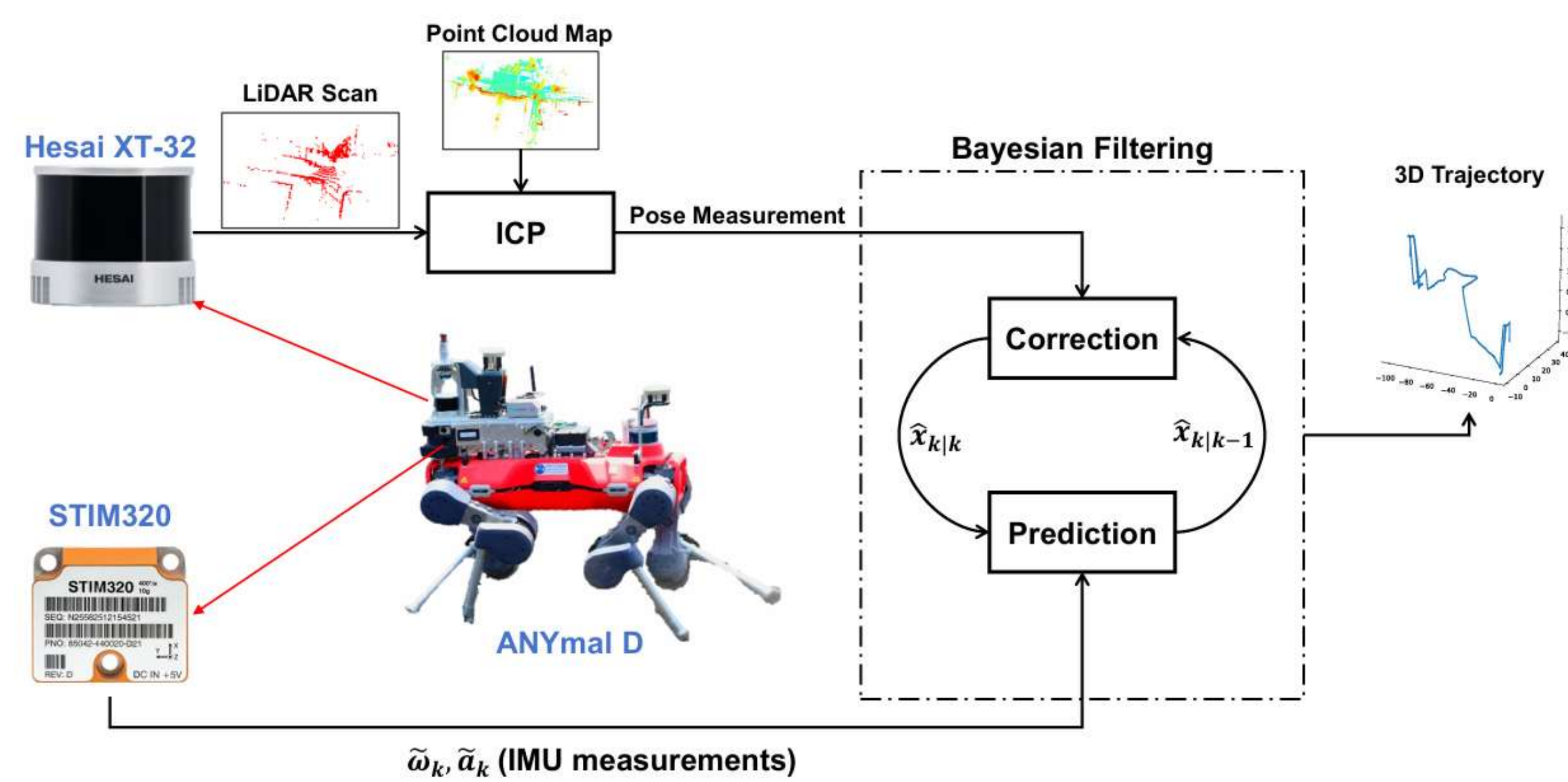
Motivation

- Accurate state estimation is a cornerstone challenge for legged platforms operating in complex environments due to terrain irregularities, foot slipping, and sensor noise.
- Need for a systematic evaluation of state estimation algorithms to determine the most robust strategy for specific constraints and environmental conditions.

Objectives

- Conduct a comprehensive benchmark of mainstream Bayesian filtering frameworks using the ANYmal GrandTour Dataset.
- Deeply investigate the impact of different approximation and sampling strategies on 3D localization performance in real-world scenarios.

Methodology



1 IMU-LiDAR State Estimation via Bayesian Filtering

- Frameworks:** Build the map, and then fuse IMU prediction with LiDAR ICP scan-to-map correction and compare four frameworks—EKF (first-order linearization), UKF (sigma-point nonlinear handling), PF (non-Gaussian, multi-modal), and InEKF (Lie-group-consistent error dynamics).

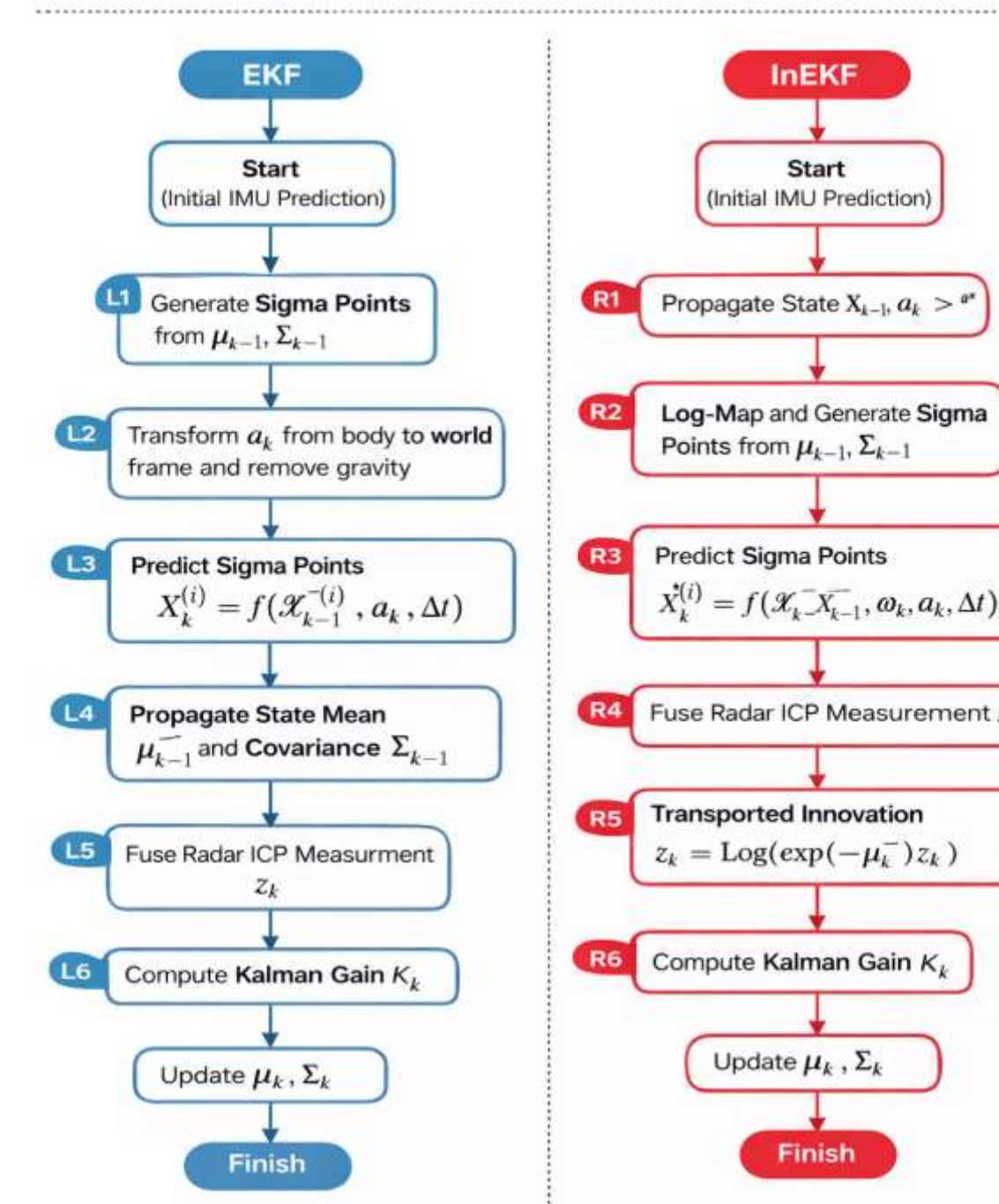
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Algorithm Implementation

Algorithm 1 PF-Based Radar-IMU Trajectory Fusion
 Require: Map $\mathcal{M}$, IMU $\mathcal{S}_{imu}$, LiDAR $\mathcal{S}_{laser}$, Extrinsic T_{li}
 Ensure: Trajectory $\mathcal{T}$
 1: Init: Load map $\mathcal{M}$; Align 1st scan via ICP; Init particles $\{x_i, w_i\}_{i=1}^N$
 2: while New LiDAR frame P_k arrives do
 3: 1. Prediction (IMU Propagation):
 4: for each particle i do
 5: Integrate IMU $\mathcal{S}_{imu}$ to update (R_i, v_i, p_i)
 6: Add process noise ϵ
 7: end for
 8: 2. Multi-Candidate ICP:
 9: Select K indices $\mathcal{J}$ (Best, Mean, Random);
 10: for each $j \in \mathcal{J}$ do
 11: $T_{ij} \leftarrow T_{li}^{-1} T_{lj}$
 12: $T_{ij}^{(0)} \leftarrow \text{ICP}(P_k, \mathcal{M}, T_{ij})$
 13: $s_j \leftarrow \text{Fitness} - \lambda \cdot \text{RMSE}$
 14: end for
 15: $(j^*, T^*) \leftarrow \arg\max(s_j)$
 16: if $\text{Fitness}(T^*) \geq \tau$ and $\text{RMSE}(T^*) \leq \tau_r$ then
 17: for each particle i do
 18: Update weight $w_i \propto w_i \cdot \mathcal{F}(s_i; T^*, \Sigma)$
 19: end for
 20: Normalize w_i
 21: end if
 22: 4. Estimation & Resample:
 23: Estimate $\hat{x} \leftarrow \sum w_i x_i$
 24: if $N_{eff} < N_{th}$ then
 25: Resample $\{x_i\}$; Apply Resampling:
 26: end if
 27: end while
 28: Store $\hat{x}$ in $\mathcal{T}$
 29: end while
 30: return $\mathcal{T}$

Algorithm 1 UKF-Based Radar-IMU Trajectory Fusion
 Require: previous state mean μ_{k-1} , previous covariance Σ_{k-1} , IMU angular velocity ω_k , IMU linear acceleration a_k , radar position measurement z_k
 1: Estimate gyro bias, initial gravity direction, and residual acceleration bias from the initial IMU segment
 2: Integrate ω_k to update the sensor orientation
 3: Transform a_k from the body frame to the world frame and remove gravity
 4: $\hat{x}_{k-1} \leftarrow$ generate sigma points from μ_{k-1} and Σ_{k-1}
 5: $w \leftarrow$ compute the corresponding sigma-point weights
 6: for each sigma point $\hat{x}_i^{(k-1)}$ do
 7: $\hat{x}_i^{(0)} \leftarrow f(\hat{x}_i^{(k-1)}, \omega_k, \Delta t)$
 8: end for
 9: $\hat{\mu}_k \leftarrow \sum_{i=0}^2 w_i \hat{x}_i^{(0)}$
 10: $\Sigma_k \leftarrow \sum_{i=0}^2 w_i (\hat{x}_i^{(0)} - \hat{\mu}_k)(\hat{x}_i^{(0)} - \hat{\mu}_k)^T + Q_k$
 11: Transform the radar ICP trajectory into the IMU frame using the static extrinsic calibration
 12: $z_k \leftarrow [p, p_r, p_l]^T$
 13: $\hat{x}_k \leftarrow$ generate sigma points from $\hat{\mu}_k$ and Σ_k
 14: $w \leftarrow$ compute the corresponding sigma-point weights
 15: for each predicted sigma point $\hat{x}_i^{(k)}$ do
 16: $\hat{x}_i^{(0)} \leftarrow h(\hat{x}_i^{(k)})$
 17: end for
 18: $\hat{z}_k \leftarrow \sum_{i=0}^2 w_i \hat{x}_i^{(0)}$
 19: $\hat{y}_k \leftarrow z_k - \hat{z}_k$
 20: $S_k \leftarrow \sum_{i=0}^2 w_i (\hat{x}_i^{(0)} - \hat{z}_k)(\hat{x}_i^{(0)} - \hat{z}_k)^T + R_k$
 21: $\Sigma_k^{(0)} \leftarrow \sum_{i=0}^2 w_i (\hat{x}_i^{(0)} - \hat{z}_k)(\hat{x}_i^{(0)} - \hat{z}_k)^T$
 22: $K_k \leftarrow \Sigma_k^{(0)} S_k^{-1}$
 23: $\hat{\mu}_k \leftarrow \hat{\mu}_k - K_k \hat{y}_k$
 24: $\Sigma_k \leftarrow \Sigma_k - K_k S_k K_k^T$
 25: return $\hat{\mu}_k, \Sigma_k$

Algorithm Flow Chart for EKF vs InEKF

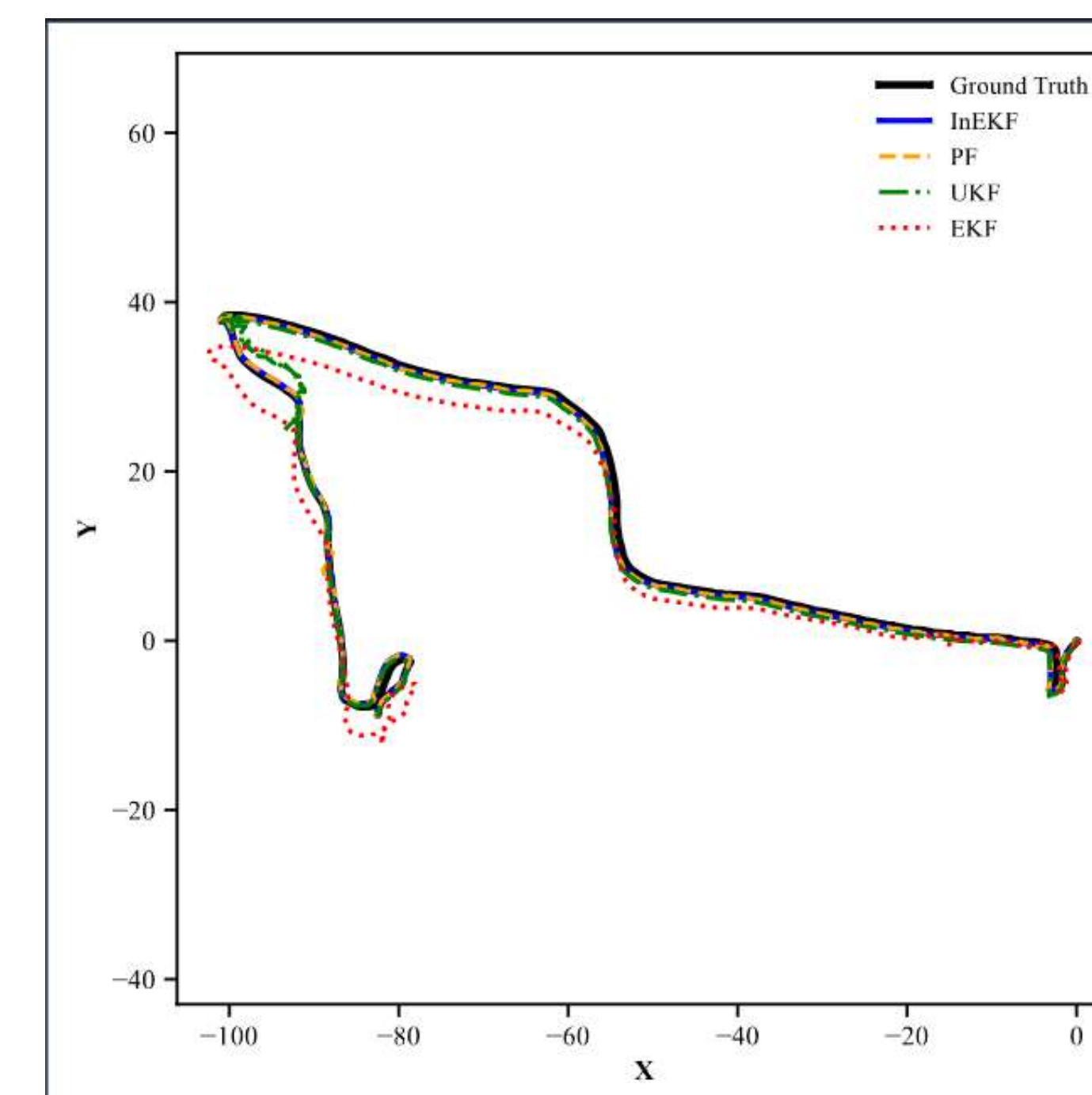


Results

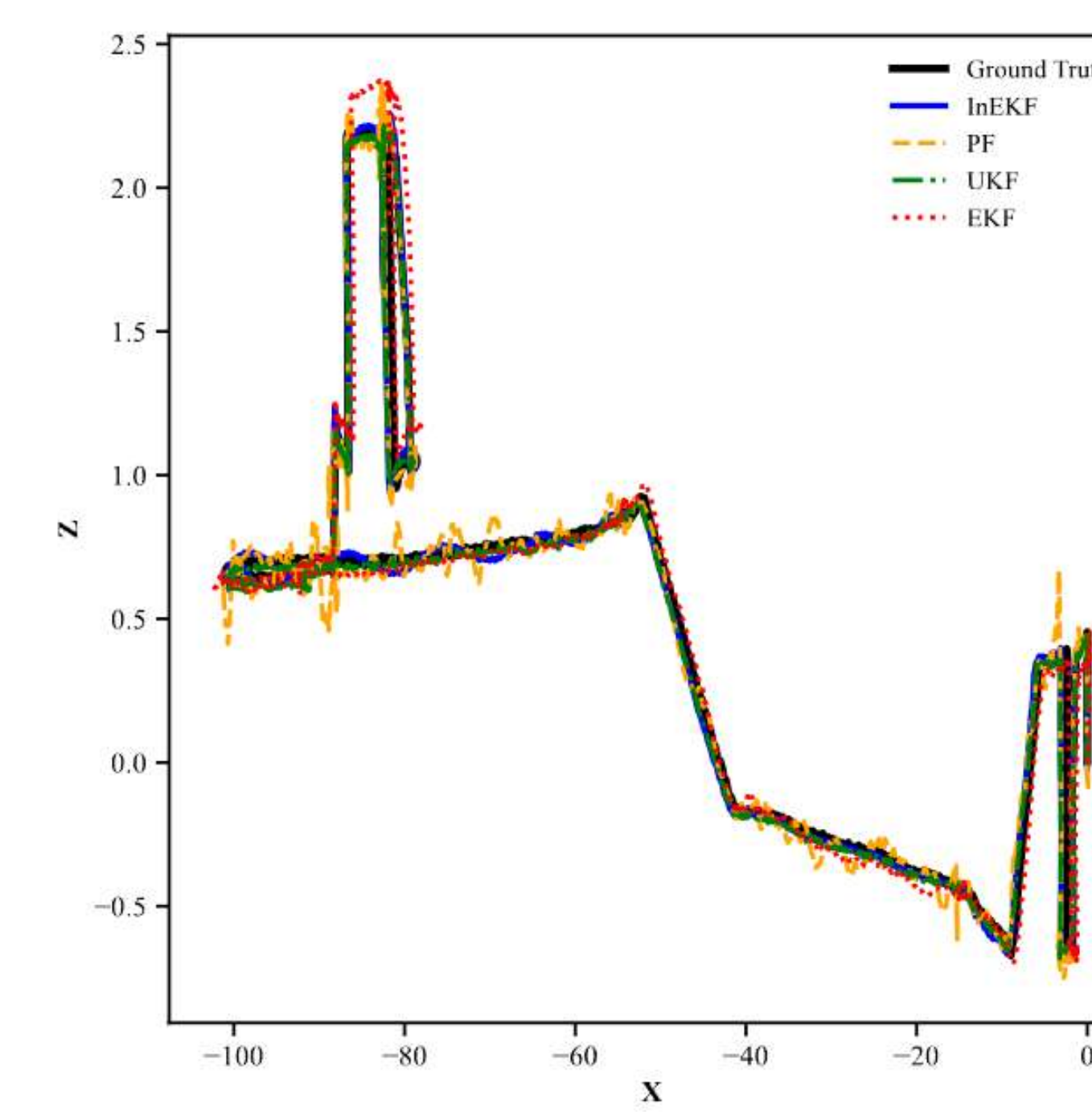
- We evaluate our method on three representative datasets (ETH-1, SPX-1, and SNOW-1) covering diverse environments, and assess localization accuracy using Absolute Trajectory Error (ATE). Here we only show the result from ETH-1 as example.
- The overall performance is quantified by the RMSE of ATE, measuring the deviation between estimated and ground truth trajectories over time.

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Trajectory Comparison



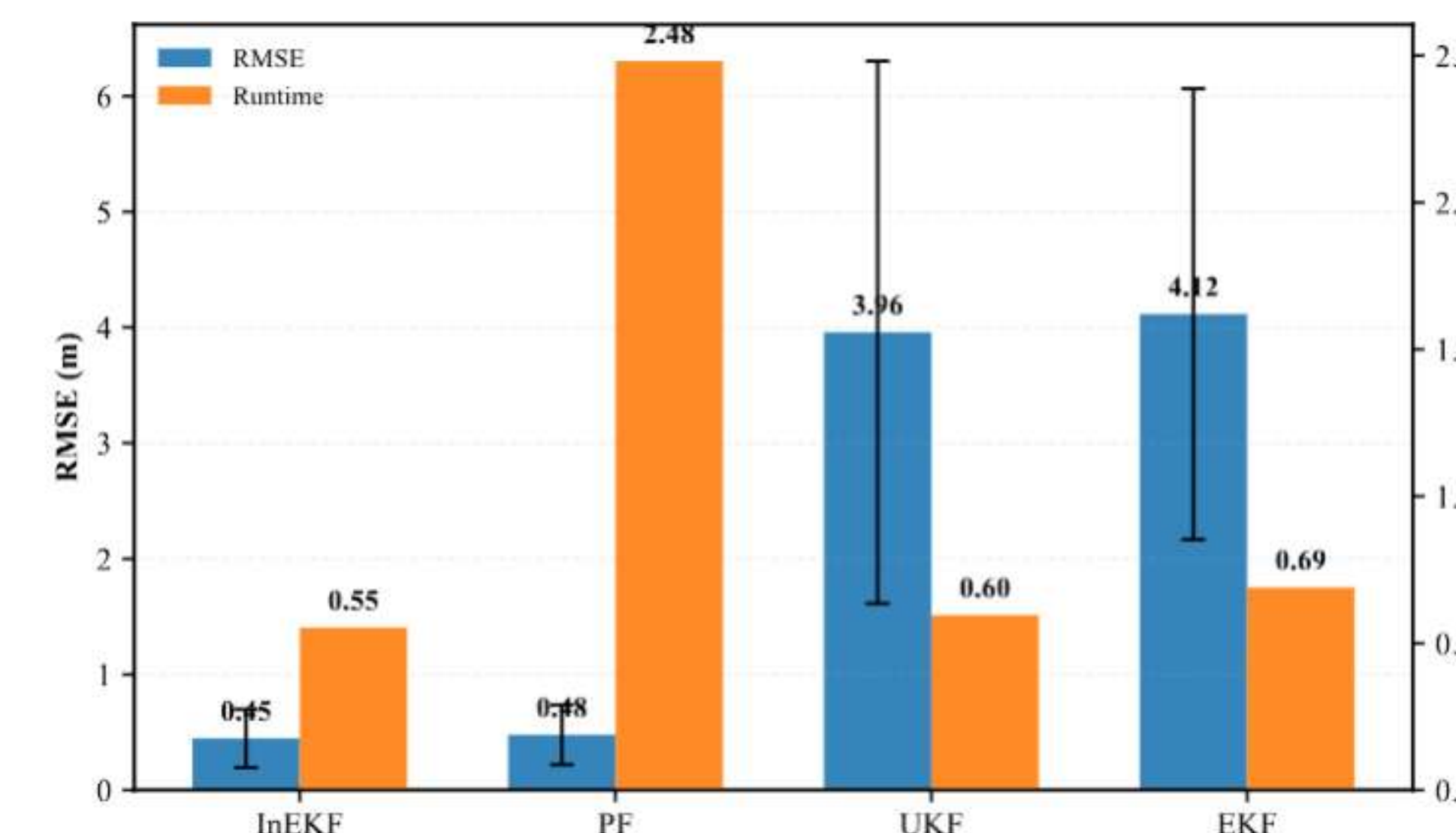
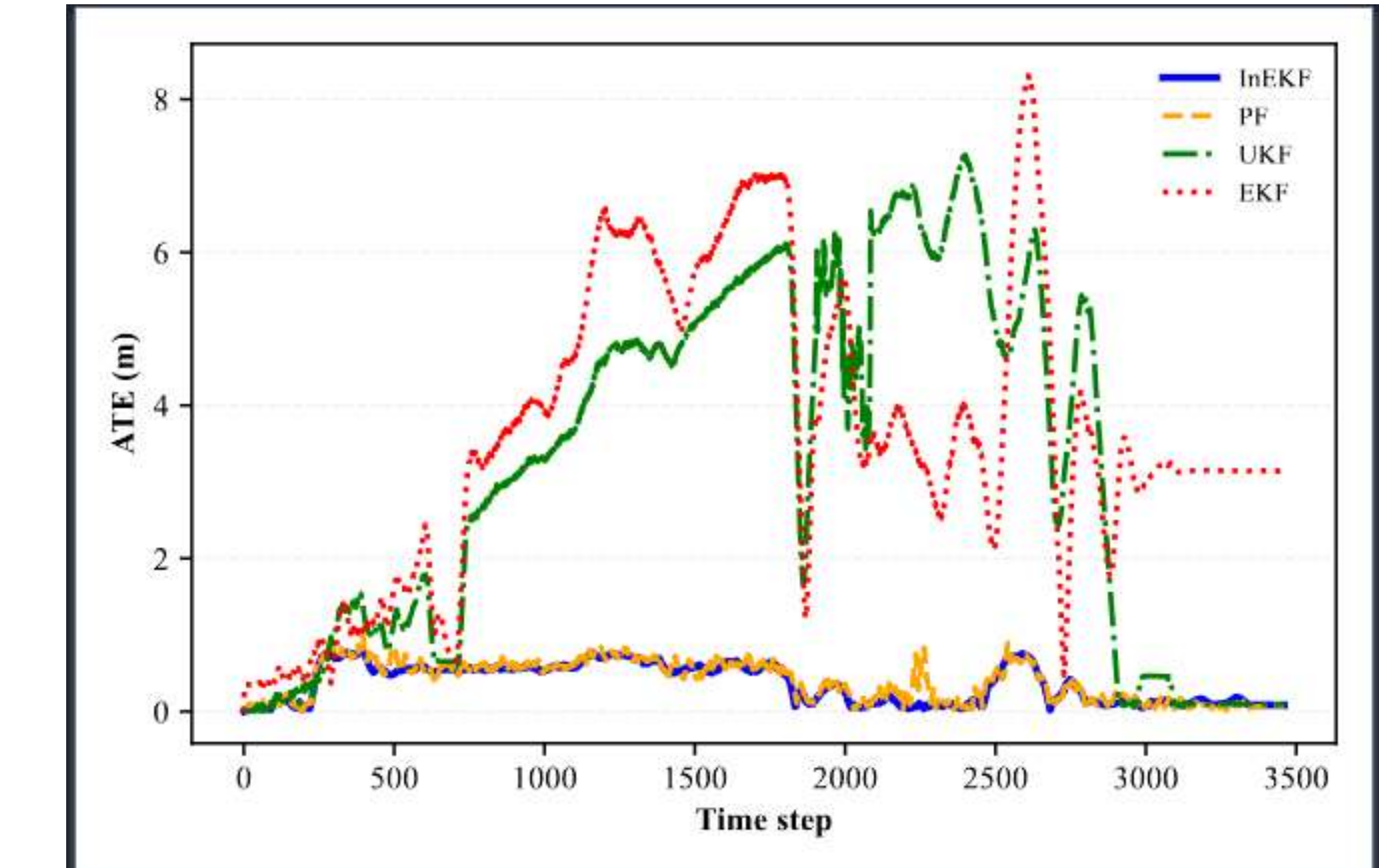
Top view



Side view

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Error and Runtime comparison



Discussion

- Our approach achieves high performance with real-time capability, especially regarding the In-EKF filter.
- All methods perform similarly in simple environments, but InEKF shows better robustness in challenging scenarios, while PF is more computationally expensive.
- The main limitation is the reliance on ICP measurements, which can lead to drift when the measurements become unreliable.

Future Works

- Future work will focus on improving robustness under perceptual degeneracy.
- Another promising direction is to incorporate contact information within an invariant filtering framework.

References: [1] J. Frey, T. Tuna, F. Fu, K. Patterson, T. Xu, M. Fallon, C. Cadena, and M. Hutter, "GrandTour: A Legged Robotics Dataset in the Wild for Multi-Modal Perception and State Estimation," arXiv:2602.18164, 2026.

[2] J. Frey, T. Tuna, L. F. T. Fu, C. Weibel, K. Patterson, B. Krummenacher, M. Müller, J. Nubert, M. Fallon, C. Cadena, and M. Hutter, "Boxi: Design Decisions in the Context of Algorithmic Performance for Robotics," in *Proc. Robotics: Science and Systems*, Los Angeles, USA, Jul. 2025.

Acknowledgment: We would like to thank Prof. Maani Ghaffari, Dr. Minghan Zhu, and the GSIs for their inspiring lectures, helpful instruction, and support during the semester and throughout this project.